

Emma Ricci-DeLucca

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I aim to understand the gap between what AI can predict from wearable data and what care providers can act on, with the goal of designing more effective human-AI collaboration in healthcare.

EDUCATION

Dartmouth College

2023 – present

PhD Candidate, Human Computer Interaction, *GPA: 3.9*

Duke University

2021 – 2023

MEng in Biomedical Engineering & Medical Device Design, *GPA: 3.8*

Swarthmore College

2017 – 2021

BS Engineering, *GPA: 3.7*

RESEARCH IN PROGRESS

DoulaConnect: AI-Enabled Health Platform

- Winner of the 2026 Dartmouth Innovation Accelerator for Digital Health pitch competition (\$40,000 award), judged by top investors and executives in digital health
- Leading front-end UX/UI design of an AI-enabled doula-facing dashboard in Figma, incorporating NLP-assisted session documentation to reduce administrative burden on community health workers
- Conducting iterative co-design sessions, prototyping, and usability testing with clinical and community stakeholders

Multimodal Sensing ML Pipeline

- Designed and evaluated multimodal Machine Learning pipelines for postpartum depression (PPD) detection using large-scale Fitbit wearable (HR/HRV, sleep metrics, activity intensity) and EHR data
- Conducted univariate analysis (Mann-Whitney U, FDR correction), multivariate logistic regression, and linear mixed-effects models to identify digital biomarkers associated with PPD
- Achieved mean AUC of 0.916 in N-of-1 PPD classification using Random Forest with SHAP-based feature attribution, identifying caloric expenditure as the top predictive biomarker

Human-Centered Qualitative User Research

- Conducted 14 semi-structured interviews with pregnant and postpartum individuals, investigating symptom recognition, care-seeking behavior, and provider communication in the perinatal period
- Applied inductive thematic analysis in Atlas.ti, iteratively coding transcripts to surface themes and synthesized findings into design implications for sensing and intervention systems

PUBLICATIONS

"I Just Need My Skin to Be Free": Understanding Consumer Wearable Experiences and Design Opportunities for Perinatal Mental Health Monitoring

Ricci-DeLucca, Batool, Jarsve, Moore, Heavey, & Murnane

Under Review, IMWUT

Digital Phenotyping of Perinatal Mental Health: A Systematic Review of Machine Learning and Wearable Sensor Approaches

Ricci-DeLucca, Heavey, & Murnane

2026, Journal of Affective Disorders Reports

Family In The Loop: Enabling Family Involvement in Dementia Care at Long-term Facilities with Person-centered AI Tools

Moore, Tao, **Ricci-DeLucca**, Tadin, Tadin, Morgan, Nottelmann, & Murnane

2025 CSCW

Wearable Biosensors for Early Detection and Continuous Monitoring of Postpartum Depression: A Human-Centered Approach

Ricci-DeLucca, & Murnane
2024 CHI Workshop Paper

Review of In-Vivo Behavior of Porous Titanium Implants for Orthopedic Applications
Ricci-DeLucca, Kelly, & Gall

2023, Journal of Medical Implants and Surgery

Gaussian curvature dilutes the nuclear lamina, favoring nuclear rupture, especially at high strain rate
Pfeifer, Tobin, Cho, Vashisth, Dooling, Vazquez, Ricci-De Lucca, Simon, & Discher

2022 Nucleus

SKILLS

Research & UX	Mixed-methdologies, qualitative research (interview, survey, focus group), thematic analysis, quantitative empirical study, usability testing
AI & ML	scikit-learn, PyTorch, mobile/wearable sensing and prediction, human-centered ML/NLP, data analysis (NumPy, pandas, R, d3.js)
Programming Design Languages	Python, R, HTML, d3.JS, Pandas, C/C++ Figma, Adobe Creative Cloud, Canva, rapid prototyping English (native), Italian (native), French (fluent)

PROFESSIONAL EXPERIENCE

Global Business Development Intern, Siemens Healthineers	Feb 2022 – Nov 2022
Developed 5 business plans for SaaS customer acquisition and co-authored a white paper supporting expansion into 15 new markets	

MENTORSHIP

Graduate Student Mentor, Dartmouth College	Sep 2024 - Present
- Mentored 7 undergraduate RAs on maternal and perinatal mental health research, recognized with Outstanding Graduate Research Mentorship Award - RAs have secured a combined \$38,600 in competitive research funding across URAD (Undergraduate Research Assistantship), ERAS (Early Research Access in the Sciences), and E.E. Just Fellowship awards	

HONORS & AWARDS

Outstanding Graduate Student Mentor, Dartmouth College	2026
Best Rural Healthcare Innovation, New England MedTech Conference	2025
Graduate Student Council Student Award, Dartmouth College	2025
Grad Cohort for Women, Computing Research Association	2025
Travel Grant, Neukom Institute for Computational Science	2024
Sensor Science, Technology and Innovation Research Traineeship, NSF	2023

SERVICE

Student Volunteer	
CHI 2026 FemTechnology Summit 2023	
Evaluator for Peer-Reviewed Journals	
IMWUT 2026 CHI 2026 CHI 2024	
Mental Health & Wellness Student Leader, Dartmouth College	2023 - 2025
Animal Welfare Volunteer & Advocate, PA, NC, NH & Italy	2020 - Present